MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
1800 SECURITY SYSTEM:FSF BASE CAMP ROADS
DESIGN CALCULATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

The Mansourah and Massarah Gold Project is in the part of the East Central Arabian Gold Region (ECAGR) within the Kingdom of Saudi Arabia, which consists of the Mansourah and Massarah deposits.

L & T Construction and Outotec, EPC Contractor of the project, contracted SETS Saudi Arabia for the Design and detail of Access Road as per MOT guidelines from Ad-Duwayhi Road to Project Location making an approx. 17.2 Km Road.

The Design of the road is based on the approved Highway Design Manual from Ministry of Transport for the Roads within Kingdom. The document is prepared for the design calculation of the access roads inside FSF BASE CAMP.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

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3. SCOPE

The Scope covers from the Topographical Survey, Geotech Investigation, Hydrology Study and design & detail of Access Road as per Highway Design Manual (HDM) by the Ministry of Transport (MOT).

The design document is prepared for the access roads inside FSF BASE CAMP.

The document is prepared based on the following input documents :

a) MD-216-4300-EG-CV-CRT-0001 (Rev C01)_ Roads Design Criteria

4. DESIGN CRITERIA & STANDARDS- ROADS

4.1. Design Speed

The selection of a proper design speed is essential because it sets curvature limits, sight distance and other geometric control aspects. Design speed is mainly dependent upon the roads hierarchy and classification. It should also take into consideration the terrain topography, anticipated operating speed, adjacent land use, and functional classification of the roadway.

Based on the consideration above and the proposed classification for MMGP Road as a rural road, the proposed design speed is 40 Kph inside FSF BASE CAMP Area.

4.2. Cross-section and Pavement Layers

As per MOT manual the main criteria of the cross-section elements for MMGP road are detailed below:

- A typical lane width of 3.65m. (Refer to section 1.06 from HDM in Annexure-B). The lane width is considered as per MOT.
- A shoulder width of 2m. (Refer to table 1.06.1 from HDM in Annexure-B). The shoulder width is considered as per MOT.
- A thickness of 0.15m for subgrade. (refer to the circular from MOT in Annexure-A)
- A thickness of 0.10m for aggregate base course. (refer to the circular from MOT in Annexure-A)
- A thickness of 0.06m for bituminous base course. (refer to the circular from MOT in Annexure-A)
- A thickness of 0.03m for bituminous wearing course. (refer to the circular from MOT in Annexure-A)

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- Design Life for the Road is 20 years
- Design Traffic Volume is 30 Vehicles (based on ESAL Range circular from MOT in Annexure A)
- Design Vehicle Type- Light passenger vehicle and Truck.
- Maximum Vehicle width- 2.6m (refer to Section F from HDM in Annexure B)
- Axle load – 9T (Refer MOT Highway Design Manual, Table 1.09.03; Assumed 12T)
- CBR – 25% (Refer doc # MD-216-49RA-EG-CV-RPT-1001: Geotechnical Investigation Report, Plant Access Road)

4.3. Horizontal Alignment

The horizontal alignment, also referred to as the road centerline, provides motorists with a facility for driving in a safe, pleasant and comfortable manner. It takes into consideration site topography, natural or man-made constraints, as well as the set design speed. MMGP Road horizontal alignment shall be developed based on the design criteria described below.

4.3.1. Sight distance

Stopping sight distance is the distance required by the driver of a vehicle traveling at a given speed to bring his vehicle to a stop after an object on the roadway becomes visible.

Passing sight distance is the minimum sight distance that must be available to enable the driver of one vehicle to pass another vehicle safely and comfortably, without interfering with the speed of an oncoming vehicle traveling at the design speed.

For the adopted Design Speed of 40 Kph on MMGP road, a minimum stopping sight distance of 45m is to be taken into account. The minimum passing sight distance is set as 280m.

4.3.2. Horizontal Curves

The design of roadway curves is based on an appropriate relationship between design speed and curvature and on their joint relationship with the super elevation and side friction.

4.3.2.1. Super elevation Rate

For every combination of curvature radius and highway design speed there is a particular rate of pavement super elevation that, in addition to the frictional force developed between the tires and the road surface, balances the centrifugal force.

The maximum Super elevation which will be used throughout the design is 6%

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4.3.2.2. Minimum Radius of curve

The minimum radius as a limiting value of curvature for a given design speed is determined from the maximum rate of super elevation and the maximum side friction factor selected for design "f".

$$R = \frac{V^2}{127 \times (0.01e + f)}$$

For the design speed of 100 Kph on MMGP road along coupled with 6% Super elevation rate the minimum radius for curve is 440m. As for speed 40 kph and a maximum super elevation of 6% the minimum turning radius is 90m.

4.3.2.3. Super elevation Transition and Attainment

The Super elevation transition consists of Tangent Run out section and Super elevation Runoff section which could be illustrated as identified herein below:

The Tangent Run out: The length of Roadway needed to change the out-side-lane slope from normal crown to Zero (flat). Refer to figure 1.05.3 from HDM in Annexure-B.

The Super elevation Runoff section: is the length of roadway needed to change the outside lane cross slope form Zero (flat) to full Super elevation and vice versa.

$$L_{runoff} = \frac{50 * e * w}{r}$$

L_{runoff} is the minimum length of super elevation runoff in (m)

r is the maximum relative gradient %, r = 0.59-0.5 (60-80Km/h) & r = 0.44(100Km/h)

W is the traveled way in (m)

e is the design super elevation rate in meter per meter

On circular curves without transition curves, two thirds of the super elevation runoff shall be placed on the tangent before the curve.

4.4. Vertical Alignment

The design of the vertical alignment of a roadway also has a direct effect on the safety and comfort of the driver. Vertical curves provide a smooth change between two tangent grades, and are designed to provide adequate stopping sight distance.

4.4.1. Tangent Grades

The minimum adopted vertical grade is 0.25% (refer to table 1.05.12 below taken from HDM). As for the maximum vertical grade for the road a 6% will be applied.

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4.4.2. Vertical Curves

The vertical curve in a vertical section or road profile is a parabolic curve (Crest Vertical Curve and Sag Vertical Curve) it is usually centered on the intersection point of the vertical tangents or PVI.

One of the principles of parabolic curves is that the rate of change of slope is constant throughout the curve. For a vertical curve, this rate is equal to the length of the curve divided by the algebraic difference of the grades. This value is called the K value and represents the distance required for the vertical tangent to change by 1 percent. The chosen K values (as per MOT manual) for the design of MMGP road are as follows:

- For the design speed of 40 kph the minimum k value for crest curves is k = 5 and for sag curves the value is k = 8

5. CONSTRUCTION OF THE ROAD

The construction of the road shall be done as per document MD-101-SMEM-EG-CV-SPC-0003.

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6. DESIGN CALCULATION OF ROAD CROSS SECTION

6.1. TRAFFIC DATA

Vehicle Type	Total Vehicle Load (t)	Single Axle load (t)	Single Axle load (kip)	No. of passes per day
Single Unit-2-Axle Truck	12	9.5	20.9	75

NB. Single Axle load, No. of passes (in above table) are reasonably assumed.

6.2. DESIGN PARAMETERS

CBR					
CBR value of subgrade		=	25 %		(MD-216-49RA-EG-CV-RPT-1001-A01(Geotechnical Investigation Report Plant Access Road))
Analysis Period					
Analysis period for road design		=	20 years		(Cl 5.2, MD-216-4300-EG-CV-CRT-0001(Roads Design Criteria))
Reliability					
Reliability	R	=	65 %		(Cl 2.13, Table 2.2,Part II,AASHTO)
Standard normal deviate	Z_R	=	-0.3885		(Table 4.1, Part I, AASHTO)
Serviceability loss					(Cl 2.2.1, Part II, AASHTO)
Initial Serviceability Index	p_o	=	4.2		
Terminal Serviceability Index	p_t	=	2.5		
Serviceability loss, ΔPSI	$p_o - p_t$	=	1.7		
Standard deviation					(Cl 2.1.3, Part II, AASHTO)
Standard deviation error	S_0	=	0.45		
Pavement layer materials characterization					(Cl 2.3.3, Part II, AASHTO)
Effective resilient modulus of -					
Asphalt Concrete	E_{AC}	=	400000	psi	
Granular Base	E_{BS}	=	30000	psi	
Granular Subbase	E_{SB}	=	15000	psi	
Roadbed material	M_R	=	12500	psi	
Layer Coefficients					
Asphalt Concrete	a_1	=	0.42		(Fig 2.5, Cl. 2.3.5, AASHTO)
Granular Base	a_2	=	0.14		(Fig 2.6, Cl. 2.3.5, AASHTO)
Granular Subbase	a_3	=	0.11		(Fig 2.7, Cl. 2.3.5, AASHTO)
Drainage coefficient					(Cl 2.4.1and Table 2.4, AASHTO)
Base material	m_2	=	1.3		
Subbase material	m_3	=	1.3		

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6.3. FUTURE TRAFFIC ESTIMATION

Estimated future traffic	w_{18}	=	D_D x D_L x $\hat{w}_{18}$	(Cl 2.1.2, Part II, AASHTO)
Directional distribution factor	D_D	=	0.5	
Lane distribution factor	D_L	=	100 %	
Traffic growth rate per year		=	1 %	(Assumed)
Traffic growth factor		=	22.15	(Table D.20, AASHTO)
ESAL Factor		=	1.83	(Table D.4, AASHTO)

Cumulative two-directional 18-kip ESAL units during the analysis period (ŵ18) shall be determined as shown below:

Design traffic (per year)	=	Current traffic x Traffic growth factor x 365
	=	606356
Cumulative Equivalent Single Axle Load (ESAL)	=	Design traffic (per year) x ESAL factor
	=	1109632
Estimated future traffic	=	D_D x D_L x $\hat{w}_{18}$
	=	554816
	=	0.55 x 10^6 ESAL

6.4. STRUCTURAL NUMBER CALCULATION

(Cl 3.1.1, Part II, AASHTO)

Structural number shall be calculated by following equation :-

$\log_{10}w_{18} = z_R S_0 + 9.36 \log_{10}(SN+1) - 0.20 + \{\log_{10}[\Delta PSI/(4.2-1.5)] / (0.40+1094/(SN+1)^{5.19})\} + 2.32 \log_{10}M_R - 8.07$

where

w_{18}	Estimated future traffic
Z_R	Standard normal deviate
SN	Structural number
ΔPSI	Serviceability loss
M_R	Effective resilient modulus of roadbed material

$z_R S_0$	=	a	=	-0.174825
$9.36 \log_{10}(SN+1) - 0.20$	=	b	=	see below table
$\log_{10}[\Delta PSI/(4.2-1.5)]$	=	c	=	-0.200915
$0.40+1094/(SN+1)^{5.19}$	=	d	=	see below table
$2.32 \log_{10}M_R - 8.07$	=	e	=	1.434831

Trial	Stability number (SN)	b	d	$\log_{10}w_{18}$	w_{18}
1	2.200	4.52820	3.01390	5.7215473	526681
2	2.240	4.57870	2.85069	5.7682282	586446
3	3.000	5.43528	1.22097	6.5307339	3394173
4	3.250	5.68172	0.99935	6.7406806	5504028
5	3.300	5.72926	0.96405	6.7808635	6037588

Calculated Structural Number SN_{cal} = 2.230

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6.5. SELECTION OF LAYER THICKNESSES

The following equation provides basis for converting SN into actual thicknesses of surfacing, base and subbase. The SN equation does not have a single unique solution; i.e., there are many combinations of layer thicknesses that are satisfactory solutions. The thickness of the flexible pavement layers is rounded to the nearest next 1/2 inch.

$SN = a_1D_1 + a_2D_2m_2 + a_3D_3m_3$

where

a_1, a_2, a_3 = Layer coefficients of surface, base and subbase course respectively

D_1, D_2, D_3 = Actual thickness of surface, base and subbase course respectively

m_2, m_3 = Drainage coefficients of base and subbase course respectively

a_1	a_2	a_3	D_1	D_2	D_3	m_2	m_3	$SN_{provided}$
0.420	0.140	0.110	2.24	3.9	5.8	1.3	1.3	2.480

Since $SN_{provided}$ > SN_{cal} OK

Hence provide following layer thicknesses -

Surface course	=	2.24	inch	=	60	mm
Granular Base Course	=	3.90	inch	=	100	mm
Granular Subbase Course	=	5.80	inch	=	150	mm

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REFERENCES

[1] Highway Design Manual by the Ministry of Transport - KSA

[2] Ministry of Transport Circular for Pavement Sections

[3] MD-216-49RA-EG-CV-TPD-1002-Topographical Survey of Road

[4] MD-216-49RA-EG-CV-RPT-1001 Geotech Investigation of Road

[5] MD-216-49RA-EG-CV-TPD-1001 Hydrology Study of Road

[6] MD-216-1000-EG-ST-CRT-1001

[7] MD-101-SMEM-EG-CV-SPC-0003

[8] MD-216-4300-EG-CV-CRT-0001

[9] MD-216-4300-EG-CV-DDE-0001

[10] MD-101-SMEM-EG-CV-CRT-0001

[11] MD-216-1000-EG-CV-MST-1012

[12] AASHTO guide for design of pavement structures (1993)

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ATTACHMENT 1 MOT's CIRCULAR FOR PAVEMENT SECTIONS

الاستخدام Application	حدود عدد المحاور المكافئة EASLsRange	الصنف Class Designation	سماكة الطبقات الإنشائية Pavement Layer thickness سم (cm)	
طرق زراعية ومغذية وطرق مفردة قصيرة	300.000 to 3 Million	خفيف L	طبقة السطح الإسفلتية	٥
			طبقة الأساس الإسفلتي	٨
			طبقة الأساس الحصوي	١٥
			طبقة القاعدة	٣٠
طرق سريعة وطرق مزدوجة وطرق محورية وطرق ذات مرور ثقيل مثل مداخل الموانئ والمدن الصناعية ؛ ومحطات الوزن وما شابهها	More than 30 Million	ثقيل جداً VII	طبقة السطح الإسفلتية	٥
			طبقة الأساس الإسفلتي	٢٠
			طبقة الأساس الحصوي	٣٠
			طبقة القاعدة	٣٠

نأمل العمل بموجبه .

ولكم تحياتنا ،،،

مدير عام إدارة الدراسات والتصميم

المهندس / عبد العني بن حامد الحربي

ATTACHMENT 2 REFERENCES FROM THE HIGHWAY DESIGN MANUAL BY MOT

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Section 1.06 Geometric Cross Section

A. Travel Lane Standards

1. Widths

No feature of a highway has a greater influence on the safety and comfort of driving than the width and condition of the surface. There is obvious need for a wide, smooth, nonskid, all-weather surface on highways. Studies on two-lane, two-way rural highways show that undesirable conditions (inadequate vehicle clearances and edge-of-pavement clearances) exist on surfaces less than 6.50 m wide carrying even moderate volumes of mixed traffic. Narrow widths severely affect capacity. Also, the effective width of the traveled way is further reduced when adjacent obstructions such as retaining walls, bridge trusses or headwalls, and parked cars restrict the lateral clearance. In addition to the capacity effect, the resultant erratic operation has an undesirable effect on driver comfort and accident rates.

Generally, studies have shown that a 7.30 m surface is required to permit desired clearance between commercial vehicles. In the interest of safety, efficiency, and ease of operation, traveled way widths of 7.30 m shall be provided on all design classes of highways except rural and urban locals. A traveled way width of 6.50 m is considered adequate for locals since meetings and passings are infrequent and the proportion of trucks is low.

Although lane widths of 3.65 m are desirable, there are circumstances that necessitate the use of lane widths less than 3.65 m. In mountainous areas with severe topography limitations and in urban areas where right-of-way and existing development become stringent controls, the use of 3.35 m lanes may be acceptable subject to MOC approval as a deviation from policy on a case-by-case basis.

Auxiliary lanes at intersections and interchanges often are provided to facilitate traffic movements. Such added lanes should be as wide as the through-traffic lanes.

Refer to 2-1.03 to determine the number of lanes required to serve the traffic on the particular roadway to be designed.

2. Cross Slope

Two-lane and urban-curbed arterial pavements on tangents or on flat curves have a crown or high point in the middle and slope downward toward both edges. On divided highways, except urban-curbed arterials, each one-way pavement shall have a unidirectional slope across the entire width of the traveled way, and the slope is usually downward to the outer edge on each pavement. Divided highways that slope in one direction are more comfortable to drivers because vehicles tend to be pulled in the same direction when changing lanes.

Roadways having a unidirectional slope may drain away from or toward the median. Drainage away from the median simplifies treatment of intersecting streets. An advantage of drainage toward the median is an economical drainage system in which all surface runoff is collected into a single conduit, and the outer lanes, which are used by most traffic, are freer of surface water. A major disadvantage of this pavement cross section is that all the pavement drainage must pass over the inner, higher speed lanes. Where curbed medians exist, drainage is concentrated next to and on the higher speed lanes. This concentration results in the annoying and hazardous splashing on the windshields of opposing traffic when the median is narrow.

The rate of cross slope on curves as well as on tangent alignment is an important element in cross section design. Pavement superelevation on curves is determined by the speed-curvature relationships given in 2-1.05, but cross slope or crown on tangents or on long radius curves is complicated by two contradictory controls. A reasonably steep lateral slope is desirable to minimize water ponding on flat sections of uncurbed pavements as a result of pavement imperfections or unequal settlement. A steep cross slope is desirable on curbed pavements to confine the flow of water to a narrow width of pavement adjacent to the curb. On the other hand, pavements with steep cross slopes are objectionable in appearance and may be annoying and uncomfortable in operation.

Steep cross slopes are undesirable on tangents because of the tendency of vehicles to drift toward the low edge of pavement. Cross slopes up to 1.5 percent are barely perceptible as far as effect on vehicle steering is concerned, but cross slopes steeper than 1.5 percent are noticeable to the driver. The latter rate requires a conscious effort in steering and would increase the susceptibility to lateral skidding when vehicles brake on wet pavements and even on dry pavements when stops are made under emergency conditions.

The prevalence of high winds may significantly alter the effect of pavement crown on steering. In rolling terrain with alternate cut-and-fill sections or in areas alternately forested and cleared, any substantial cross wind produces an intermittent impact on a vehicle moving along the highway and affects its steering. In

sult highway maps, to rest, or for other personal reasons.

- Width is provided for increased mobility to escape potential accidents or reduce their severity.
- The sense of openness created by shoulders contributes much to driving ease and freedom from strain.
- Sight distance is improved in cut sections, thereby improving safety.
- Some types of shoulders enhance the aesthetics of the highway.
- Highway capacity is improved, and uniform speed is encouraged.
- Space is provided for maintenance operations.
- Lateral clearance to signs and guardrails is increased.
- Storm water can be discharged farther from the pavement, and seepage adjacent to the pavement can be minimized. This may reduce pavement breakup.
- Structural support is given to the pavement.
- Space is provided for pedestrian and bicycle use.
- Space is provided for bus stops.

Pavement edge dropoff, which can cause loss of vehicle control, is eliminated when the shoulder is stabilized.

Where on-street parking is permitted, the parking lane provides some of the same services listed above for shoulders.

2. Widths

Table 1.06.1 shows the desirable and minimum shoulder widths for the eight major classes of roadways.

Table 1.06.1
Shoulder Widths (m)

Class	Special Feature	Desirable	Minimum
Rural Local	Flat	2.00	Not Applicable
	Rolling	2.00	Not Applicable
	Mountainous	2.00	1.25
Rural Collector	Flat	2.50	Not Applicable
	Rolling	2.50	Not Applicable
	Mountainous	2.50	1.25
Rural Arterial and Expressways	Flat	Inside Sh. 2.00	Not Applicable
		Outside Sh. 3.00	Not Applicable
	Rolling	Inside Sh. 2.00	Not Applicable
		Outside Sh. 3.00	Not Applicable
	Mountainous	Inside Sh. 2.00	Not Applicable
		Outside Sh. 3.00	2.50
Urban Local	Does Not Apply—Parking Lanes Are Required		
Urban Collector		2.00	1.25
Urban Arterial		Inside Sh. 2.00	Not Applicable
		Outside Sh. 3.00	2.50
Urban Expressways		Inside Sh. 2.00	Not Applicable
		Outside Sh. 3.00	Not Applicable

Note: For climbing lanes, the outside shoulder width shall be 1.25 m for all classes of roads.

Section 1.05 Geometric and Structure Standards

A. Sight Distance

1. General

Sight distance is the continuous length of roadway ahead, visible to the driver. Of utmost importance in highway design is the arrangement of geometric elements so adequate sight distance exists for safe and efficient operation. Two basic types of sight distance must be considered; that required for stopping and that required for passing overtaken vehicles. Safe stopping sight distance must be provided continuously on all highways. Safe passing sight distance is applicable only on two-lane highways, primarily in rural or in outlying urban areas. Sight distance for intersections at grade including railroad crossings is covered under 2-1.07.

2. Stopping Sight Distance

Stopping sight distance is the distance required by the driver of a vehicle traveling at a given speed to bring his vehicle to a stop after an object on the roadway becomes visible.

The distance required for stopping under prevailing vehicle and climatic conditions depends on the initial speed of the vehicle, the perception and reaction time of the driver, and the coefficient of friction between the tires and the roadway. The minimum stopping sight distance is computed using the following equation:

$$S_s = \frac{Vt}{3.6} + \frac{V^2}{255f}$$

where S_s = Minimum stopping sight distance in meters
V = Vehicle running speed in kilometers per hour
t = Combined perception and reaction time in seconds
f = Coefficient of friction

The minimum stopping sight distance values for various design and running speeds are shown in Table 1.05.1. These values have been rounded higher to the nearest 5 m and are based on stopping on level grades, a perception and reaction time of 2.5 s, and coefficients of friction for wet pavements.

3. Passing Sight Distance

Passing sight distance is the minimum sight distance that must be available to enable the driver of one vehicle to pass another vehicle safely and comfortably, without interfering with the speed of an oncoming vehicle traveling at the design speed.

It is important, for reasons of safety and service, to provide as many passing opportunities on two-lane highways as possible. The passing sight distance available on a section of highway influences the average speed of the traffic, particularly when a highway is operating near capacity.

Minimum values for passing sight distance are shown in Table 1.05.1.

4. Sight Distance Standards

Table 1.05.1 shows the standards for stopping and passing sight distances for various design and assumed running speeds. These are minimum values, and wherever possible, larger values should be used.

Table 1.05.1
Sight Distance Standards

Stopping Sight Distance				
Design Speed (km/h)	Assumed Running Speed (km/h)	Coefficient Of Friction	Minimum Stopping Sight Distance S_s (m)	Minimum Passing Sight Distance S_p (m)
40	4	0.38	45	280
50	50	0.36	65	340
60	60	0.34	85	420
70	70	0.32	110	480
80	80	0.31	140	560
90	90	0.30	170	620
100	100	0.30	200	680
110	109	0.29	240	740
120	118	0.28	280	(a)800
130	125	0.27	315	(a)860
140	128	0.27	330	(a)920

(a) Applicable only to highways with full control of access.

Note: The minimum stopping sight distances should be increased for sustained downgrades as indicated in Table 1.05.2.

Table 1.05.2
Effect of Grade on Stopping Sight Distance

Design Speed (km/h)	Increase Stopping Sight Distance for Downgrades (m)		
	3%	6%	9%
40	2	4	6
50	3	6	10
60	5	10	18
70	7	15	26
80	9	21	(a) —
90	12	29	—
100	16	38	—
110	20	48	—
120	24	(a) —	—
130	29	—	—
140	35	—	—

(a) Design Condition

5. Measurement of Sight Distance

Sight distance is measured from the driver's eyes, which are assumed to be 1.05 m above the pavement surface to an object on the roadway ahead. For stopping sight distance the object is assumed to be 15 cm high. For passing sight distance the object is assumed to be 1.30 m high.

6. Sight Distance for Crest Vertical Curves

The minimum length of a crest vertical curve is controlled by stopping sight distance, or if passing is desired, by passing sight distance requirements. The minimum vertical curve length is expressed by the following equation:

$$L = KA$$

where L = Length of vertical curve in meters
K = Expression of vertical curvature (constant for each design speed)
A = Algebraic difference in percent of gradient

The crest vertical curvature constant for stopping sight distance conditions is computed by the equation:

$$K_{sc} = \frac{S_s^2}{405}$$

where K_{sc} = K Value for stopping sight distance at vertical crests
S_s = Minimum stopping sight distance

The crest vertical curvature constant for passing sight distance conditions is computed by the equation:

$$K_{pc} = \frac{S_p^2}{943}$$

where K_{pc} = K Value for passing sight distance at vertical crests
S_p = Minimum passing sight distance

Values for K for various design speeds are shown in Table 1.05.3.

Table 1.05.3
Crest Vertical Curve "K" Values

Minimum Vertical Curvature		
	"K" Values	
Design Speed (km/h)	Stopping Conditions K_{sc}	Passing Conditions K_{pc}
40	5	83
50	11	123
60	18	188
70	30	244
80	49	333
90	72	408
100	99	490
110	143	581
120	194	679
130	245	784
140	269	898

The minimum length of crest vertical curve determined using $L = KA$ should be rounded higher to an even 10 m.

7. Stopping Sight Distance for Sag Vertical Curves

The minimum stopping sight distance for a sag vertical curve is controlled by stopping sight distance, which in turn, is controlled by the vehicle headlight sight distance. The sag vertical curvature must be flat enough so that the headlights illuminate the roadway ahead for a distance of at least the stopping sight distance.

It is assumed that the headlight beam slopes upward at an angle of 1^G from the plane of the vehicle and that the height of the headlights is 0.6 m above the roadway. The minimum length of vertical curve for the above conditions is expressed by the equation:

$$L = KA$$

where the terms of the equation are as in Section 2-1.05A6. The sag vertical curvature constant is expressed by the equation:

$$K_{ss} = \frac{S_s^2}{122 + 3.5S_s}$$

where K_{ss} = K Value for stopping sight distance at vertical sags
S_s = Stopping sight distance

Values for K for various design speeds are shown in Table 1.05.4.

Table 1.05.4
Sag Vertical Curve "K" Values
For Stopping Sight Distance

Design Speed (km/h)	Minimum Vertical Curvature "K" Value K_{ss}
40	8
50	12
60	18
70	24
80	32
90	41
100	49
110	60
120	71
130	81
140	86

For aesthetics and driving comfort, sag vertical curve lengths determined from the above criteria should be increased substantially if economically feasible.

8. Sight Distance on Horizontal Curves

Where an obstruction on the inside of a horizontal curve restricts sight distance, the minimum radius of curvature is determined by sight distance, either stopping or passing, as applicable. Some examples of obstructions are bridge piers, buildings, signs, and cut slopes. The driver's line of sight to the roadway ahead is intercepted by the obstruction. For stopping sight distance, the height of this intercept is assumed to be 0.60 m above the center of the inside lane (inside with respect to the horizontal curve). The intercept height for passing sight distance is assumed to be 1.17 m. These intercept heights are for tangent grades and must be adjusted up or down for vertical curvature.

The clearance, C, is the distance from the center of the inside lane to a point where the sight line intercepts the obstruction. Figure 1.05.1 shows the relations between clearance, radius, design speed and stopping sight distance. Figure 1.05.2 shows the same information for passing sight distance. To increase the clearance, the obstruction should be moved away from the roadway, or possibly eliminated, or else the alignment must be revised.

9. Sight Distance Through Underpasses

Stopping sight distance must be maintained. If economically feasible, passing sight distance shall also be maintained as the highway passes under a crossing structure.

B. Superelevation

1. Basic Criteria

As a vehicle traverses a horizontal curve, centrifugal force is opposed by roadway superelevation and by side friction between the tires and the roadway surface. The minimum allowable radius for any design speed can be computed using the following equation:

$$R = \frac{V^2}{127\,(e+f)}$$

where R = Minimum radius of circular curve in meters
V = Vehicle speed in kilometers per hour
e = Maximum superelevation rate in meters per meter
f = Side friction factor

Side friction factors for wet pavements are used for highway design. The maximum safe values of side friction factors for various design speeds are shown in Table 1.05.5.

Table 1.05.5
Maximum Safe Side Friction Factors

Design Speed (km/h)	Side Friction Factor
40	0.17
50	0.16
60	0.15
70	0.15
80	0.14
90	0.13
100	0.12
110	0.11
120	0.10
130	0.09
140	0.08

Minimum radii of curves for various design speeds and maximum rates of superelevation are shown in Table 1.05.6. The radii have been rounded to the nearest 5 m.

Table 1.05.6
Minimum Radii for Superelevated Curves

Design Speed (km/h)	Maximum Superelevation Rate (m/m)			
	0.06	0.08	0.10	0.12
40	55	50	45	45
50	90	85	75	70
60	135	125	115	105
70	185	170	155	145
80	250	230	210	195
90	335	305	280	255
100	440	395	360	330
110	560	500	455	415
120	710	630	565	515
130	885	785	700	635
140	1,100	965	860	770

2. Superelevation Rates

The maximum superelevation rates for various highway design classifications are shown in Table 1.05.7.

Table 1.05.7
Maximum Superelevation Rates

Highway Design Class	Desirable Maximum Superelevation Rate (m/m)	Absolute Maximum Superelevation Rate (m/m)
Expressway	0.08	0.10
Arterial	0.08	0.10
Collector	0.08	0.12
Local	0.10	0.12

Lower superelevation rates for all classifications of highways may be necessary in urban areas where restricted speed zones or at-grade intersections are controlling factors. In addition, established street grades, curbs, or drainage may impose limitations on design. Such conditions may warrant, for example, a reduction in the superelevation rate, different rates for each half of the roadbed, or both. In warping street cross sections for drainage, adverse

superelevation should be avoided. A superelevation rate of not more than 0.06 m/m shall be used for urban arterial highways with the above conditions.

A maximum superelevation rate of 0.08 to 0.12 m/m is used for ramps and turning roadways associated with interchanges and at-grade intersections.

3. Axis of Rotation

a. Undivided Highways. The axis of rotation for superelevation shall be at the centerline of the roadbed. However, in special cases such as roads in flat country where curves are preceded by long, relatively level tangents, the plane of superelevation should be rotated about the inside edge of the traveled way. This will improve perception of the curve and avoid drainage pockets.

b. Ramps and Expressway Connections. The axis of rotation may be either the edge of the traveled way or the centerline. Appearance and drainage considerations shall be taken into account when selecting the axis of rotation.

c. Divided Highways. The axis of rotation shall be at the median edge of each traveled way, except as follows: To avoid a sawtooth effect on bridges with decked-medians, the axis of rotation shall be at the centerline. The selection of the axis of rotation shall always be considered in conjunction with the design of the profile grade and superelevation transition.

4. Superelevation Development

The superelevation development shall be designed to satisfy the requirements of safety and comfort and should be pleasing in appearance. Figure 1.05.3 illustrates the desirable methods for developing superelevation. When transition curves are used, the transition curve and superelevation runoff should be coincident. On circular curves without transition curves, two-thirds of the superelevation runoff shall be placed on the tangent before the curve. The length of the superelevation runoff shall be long enough so the runoff slope meets the criteria shown in Table 1.05.8. The runoff slope is in longitudinal gradient between the edge of the traveled way profile and the center of the traveled way profile. The minimum length of superelevation runoff can be computed using the following equation:

$$L = \frac{50\,ew}{r}$$

where L = Minimum superelevation runoff length in meters
e = Superelevation rate in meters per meter
w = Width of traveled way in meters
r = Percent of runoff slope

For example, a two-lane highway has a design speed of 80 km/h, a traveled way width of 7.3 m, and a superelevation rate of 0.10 m/m. The minimum length of superelevation runoff should be 73 m. To simplify design, runoff lengths should be selected to be compatible with the whole numbered transition curve parameter, A, and radius, R.

Table 1.05.8
Superelevation Runoff Slopes

Design Speed (km/h)	Runoff Slope Percent "r"
40	0.74
50	0.66
60	0.59
70	0.54
80	0.50
90	0.47
100	0.44
110	0.41
120	0.38
130	0.36
140	0.34

For expressway and arterial highways, the minimum length of superelevation runoff shall be 50 m. For collector and local highways, the minimum shall be 30 m.

For long transition curves where the runoff slope, r, is less than 0.30 percent, the attainment of superelevation within the superelevation runoff as shown in Figure 1.05.3 should be modified as follows:

a. From section B (level on the high side, normal crown on the low side) to section C (superelevation at the normal crown rate), the runoff slope should be 0.30 percent.

b. From section C to section D (full superelevation), the runoff slope should be less than 0.30 percent.

In special restrictive situations where the standard superelevation rate is not feasible or the desirable runoff length is not attainable, the highest possible rate and longest length respectively shall be used. These situations occur on interchange ramps, frontage roads, and separate turning lanes where the length of radius, curve, and adjacent tangents are short.

After the superelevation development is designed, profiles of edges of the traveled way should be plotted and irregularities removed by introducing smooth curves. Flat areas that may not drain properly must be avoided.

5. Superelevation of Broken Back Curves

Broken back horizontal alignment occurs

F. Design Vehicles

The turning radii and other dimensions shall be large enough to meet the turning requirements of vehicles using the intersection. Four types of design vehicles with their dimensions and turning paths are shown in Figures 1.07.5, 1.07.6, and 1.07.7. Intersections shall be designed to accommodate at least a single unit truck. If it is anticipated that trailer truck combinations will be using the highway, the turning requirements for WB-12 or WB-15 trailer truck shall be met.

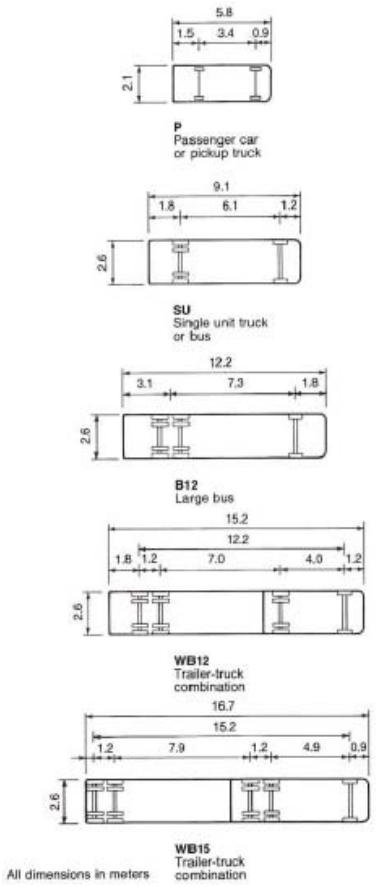
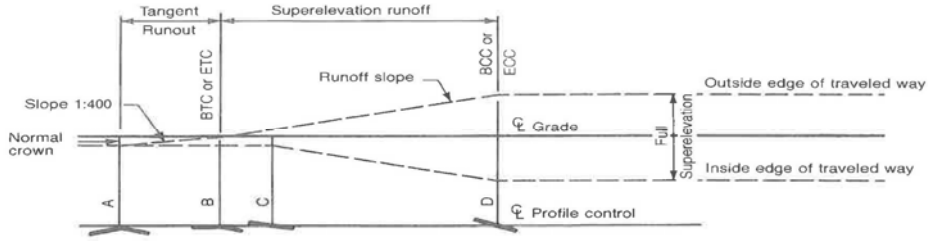
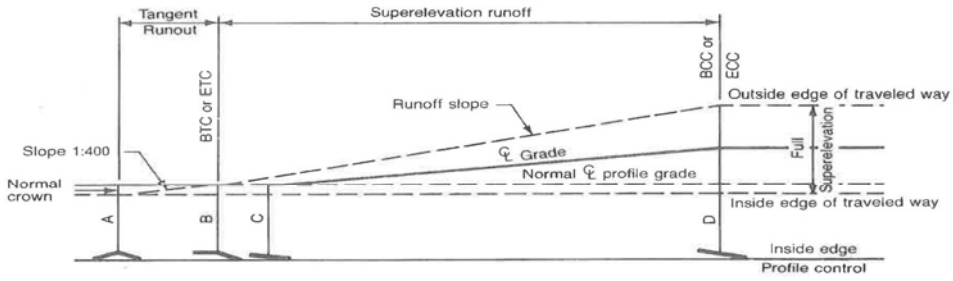


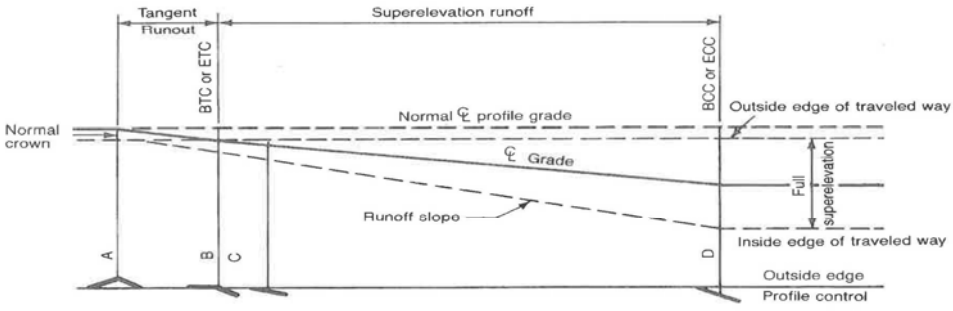
Figure 1.07.5
Design Vehicles



Traveled way revolved about centerline



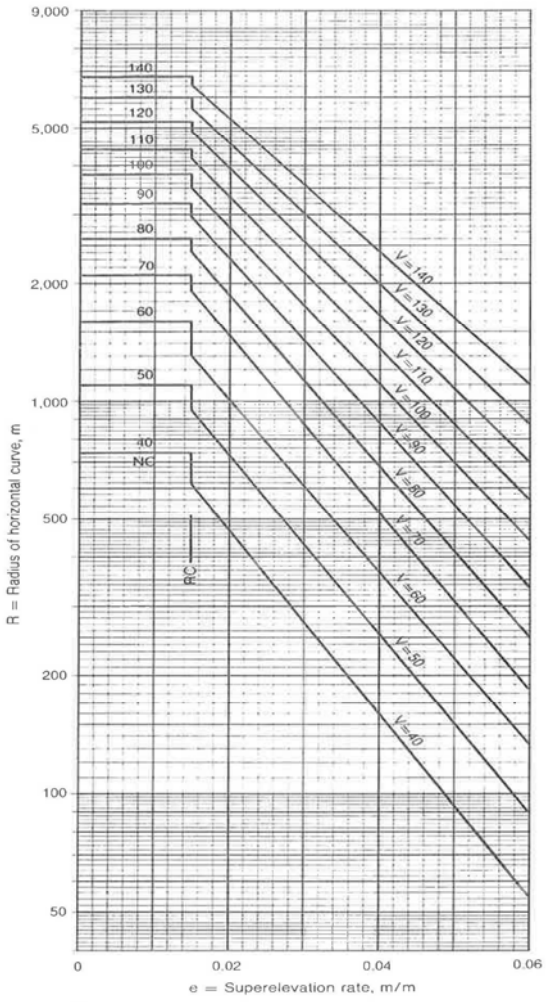
Traveled way revolved about inside edge



Traveled way revolved about outside edge

Notes:
A = Normal crown,
B = Level highside, normal crown low side
C = Superelevation at normal crown rate
D = Full superelevation

Figure 1.05.3
Development of Superelevation



Notes:
V = Design speed, km/h
RC = Remove crown and superelevate at normal crown rate
NC = Normal crown, superelevation not required

Figure 1.05.4
Design Superelevation Rates
$e_{Max.} = 0.06$

Table 1.05.11
Maximum Percent of Gradients

Design Speed (km/h)	Highway Design Class							
	Expressway		Arterial		Collector		Local	
	Desirable	Absolute	Desirable	Absolute	Desirable	Absolute	Desirable	Absolute
40							8	12
50					7	9	7	11
60					6	8	6	10
70			4.5	6	5	6	5.5	9
80	4	5	4	5	4	7	5	8
90	3.5	5	3.5	5	3.5	6.5	5	7
100	3	5	3	5	3	6		
110	2.5	4.5	2.5	4.5				
120	2.5	4	2.5	4				
130	2	3.5						
140	2	3						

In areas where the surface water is above the ground level or the ground water table is immediately below the surface, the grade line shall be established so that the low edge of the finished shoulder is at least 0.5 m above the water level. In areas of rock, if practical, the grade line should be established so that the low edge of the finished shoulder is at least 0.3 m above the rock level. This will avoid unnecessary rock excavation.

For aesthetic reasons, the length of vertical curves should be substantially longer than the length required for stopping sight distance.

2. Grade Standards

Maximum gradients for various highway design classes and design speeds are shown in Table 1.05.11.

Minimum longitudinal gradients for highways and ditches are shown in Table 1.05.12.

Table 1.05.12
Minimum Percent of Gradients
For All Design Classes

Design Element	Gradient	
	Desirable Min. (%)	Absolute Min. (%)
Cuts or Fills		
Curbed	0.5	0.25
Uncurbed	0.5	0.5
Flat Bottomed Ditches	0.5	0.05
"V" ditches	0.5	0.25

3. Position with Respect to Cross Section

The elevation of the pavement, subgrade, and other features of the highway are established by the grade line.

In relation to the cross section, the horizontal position of the grade line should be as follows:

a. Undivided Highways

The grade line should coincide with the highway centerline.

b. Ramps and Directional Highway Connections

The grade line may be positioned at either edge of the traveled way or at the centerline.

c. Divided Highways

The grade line may be positioned at either the centerline of the highway or at the median edge of the traveled way.

4. Vertical Curves

Properly designed vertical curves will provide adequate sight distance, safety, comfortable driving, good drainage and pleasing appearance.

A parabolic curve is used. Figure 1.05.11 shows the different types of vertical curves and gives the necessary mathematical relations for computing vertical curves.

For design speeds of 60 km/h or more, the minimum length of vertical curve shall be 100 m. For design speeds of 50 km/h and below, it shall be 50 m whenever feasible. Vertical curves with curvature that is too flat may develop poor drainage. This difficulty may be overcome by a slight gutter or ditch grade adjustment or by shortening the vertical curve.

In special cases, unsymmetrical vertical curves may be used provided that sight distance and other criteria set forth herein are not violated.

5. Heavy Grades

Except in level country, it is often not economically feasible to design a profile grade that will allow uniform operating speeds for all classes of vehicles. Long sustained grades are sometimes unavoidable.

Where guardrails, walls, or other vertical elements are used, the graded shoulder should be wide enough so that these vertical elements can be offset a minimum of 0.5 m from the outer edge of the usable shoulder. It may be necessary to provide a graded shoulder wider than used elsewhere to provide lateral support for guardrail posts. Shoulders on structures should have the same width as the usable shoulders on the approach roadways, both right and left. Refer to 2-1.15 for widening required for traffic barriers.

3. Cross Slopes

Cross slopes for shoulders shall be 4 percent except that the inside shoulders on multilane facilities shall be 1 percent sloping toward the median.

D. Side Ditches

The ditch cross sections must have adequate hydraulic capacity. The flow line of ditch must be adequate to accommodate drainage of the pavement.

Ditches should have a streamlined cross section for safety and ease of maintenance. See 2-1.15C2. Wide ditch bottoms facilitate grading operations where side borrow methods are used.

The depth of the ditch in cut sections is dependent on hydraulic needs but should normally be no deeper than 0.25 m below subgrade for safety and maintenance. Short sections of ditch deeper than 0.25 m may be required to drain flat gradient ditches to a culvert inlet or to an embankment area. Should hydraulic needs dictate ditches of greater capacity than the normal depth V-ditch, the designer should design a flat bottom ditch of sufficient width instead of deepening the V-ditch. Subsurface ground water may require the installation of underdrain systems in lieu of deeper or wider ditches.

E. Sideslopes

Sideslopes should be designed to insure the stability of the roadway and to provide a reasonable opportunity for recovery for an out-of-control vehicle.

The desirable foreslope, the slope from the edge of the surfaced shoulder to the edge of the subgrade shoulder, is 6:1 except that on rural local roads with an ADT (20) less than 300, the foreslope shall be 4:1. In excavation sections with ditches, the slope from the edge of the subgrade shoulder to the bottom of the ditch shall be an extension of the foreslope.

Earth cut and fill slopes should be flattened as appropriate with the topography and consistent with the overall type of highway. Effective erosion control, low-cost maintenance, and adequate drainage of the subgrade are dependent upon proper shaping of the sideslopes. Slope and soil data are used in combination to approximate the stability of the slopes and the erosion hazard potential. Overall economy depends not only on the element of first cost but also on the cost of maintenance, of which slope stability is a factor. In addition to these reasons for flat slopes on any highway, the proximity of any urban highway to the developments and people of the community call for additional attention to naturalized slopes and overall appearance factors.

Earth cutslopes shall not be steeper than 2:1 and preferably flatter, at least 3:1, to make it easier for motorized equipment to be used in maintenance in areas where rock or other special soils are encountered. See HDM-1-2.06 for recommendations in rock. Retaining walls of partial height shall be considered where space restrictions require slopes that otherwise would be steeper than 2:1. On the other hand, soil characteristics may necessitate the use of slopes flatter than 2:1 or even 3:1. If adequate width is not available in such cases, partial walls may be necessary. To minimize the feeling of constriction, walls should be set back a minimum of 0.5 m from the outer edge of the usable shoulder.

The highway should be designed with a clear zone free of hazards. Desirably, slopes should be 6:1 or flatter. If slopes are steeper than 3:1, a traffic barrier may be required. The purpose and width of clear zone and warrants for traffic barriers are described in 2-1.15.

The intersections of slope planes in the highway cross section should be well rounded to simulate natural earth forms.

The combination of flat slopes and rounding is frequently referred to as streamlined cross section. With this shape the cross winds sweep along the surface without forming eddies that contribute to the wind erosion and drifting of sand. The streamlined cross section usually results in a minimum of expenditure for sand removal because the winds blow the sand off the pavement instead of drifting it, as happens in cross sections with steep slopes without rounding. When combined with the design practice of making the level of the roadway above that of the surrounding land to also insure drainage of the subgrade, the streamlined cross section results in a highway requiring a minimum of maintenance and operating costs and a maximum of safety. See Figure 1.06.5 and 2-1.16 for special cut-and-fill treatment in sand.

thetic treatment. For maximum efficiency, a median should be highly visible both night and day and in definite contrast to the through-traffic lanes. A median may vary in scope from a barrier to an area of varying width between two independently designed roadways. Medians may be depressed, raised, or flush with the pavement surface. Medians shall be as wide as feasible but of a dimension in balance with other components of the cross section. As far as the safety and convenience of motor vehicle operation are concerned, the farther the traveled ways are apart, the better. However, economic factors limit the width of median that can be provided.

Cost of construction and maintenance increases generally with an increase in the width of roadbed, but the additional cost may not be appreciable compared with the cost of the highway as a whole and may be justified in view of the benefits derived. A distinct advantage of the wider medians on roadways other than expressways is to provide adequate shelter for vehicles crossing at intersections with public highways and at crossovers serving commercial and private drives. However, wide medians are a disadvantage when the intersection is signalized. The increased time for vehicles to cross the median may lead to inefficient signal operation.

If the right-of-way is restricted, the median shall not be widened beyond a desirable minimum at the expense of narrowed border areas. A reasonable border width is required to adequately serve as a buffer between the private development along the highway and the traveled way, particularly where zoning is limited or nonexistent. Space must be provided on the borders for sidewalks, highway signs, utility lines, parking, drainage channels and structures, for proper slopes, and any retained native growth. Narrowing these areas may tend to develop hazards and hindrances similar to those that the median is designed to avoid.

A depressed median is preferred on expressways. This type is more efficient for drainage. Median side slope should be no steeper than 6:1, and preferably 10:1. Slopes of 4:1 may be adequate structurally and for vegetation, but they are questionably steep for roadside safety. All drainage inlets in the median shall be designed with the top flush with the ground or culvert ends provided with safety grates so they will not be hazardous to out of control vehicles that run off the roadway.

Raised medians have application on urban streets where it is desirable to regulate left-turn movements. They are also frequently used where the median is to be landscaped, particularly where the width is relatively narrow. It must be pointed out, however, that landscaping in narrow medians creates hazardous conditions for maintenance operations and may also restrict sight distance on curves.

Flush medians are used to some extent on all types of urban arterials. When used on expressways, a median barrier may be required. The median should be slightly crowned or depressed for drainage. The crowned type is frequently used because it eliminates the need for collecting drainage water in the median. However, the slightly depressed type is generally preferred with the depression being formed by 4-percent cross slopes being constructed from the shoulder of each inside lane and intersecting in the middle of the median.

2. Median Widths

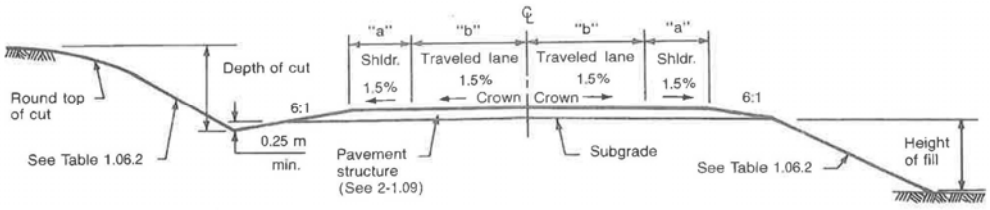
The desirable median width is 20 m for all rural and urban highways except that the desirable width for rural mountain highways is 10 m.

Under restraining circumstances that arise due to economical, topographical, environmental, or political factors the median width may be reduced to a minimum width of 5 m for any class of highway.

Whenever two abutting highway sections have different median widths, a smooth transition shall adjust the difference. For good appearance the transition length shall be as long as possible and occur within a horizontal curve.

G. Cross Sections

Figures 1.06.1, 1.06.2, 1.06.3, and 1.06.4 show cross sections depicting the essential elements for typical sections of two-lane rural highways, urban streets, multilane rural highways, and urban highways.



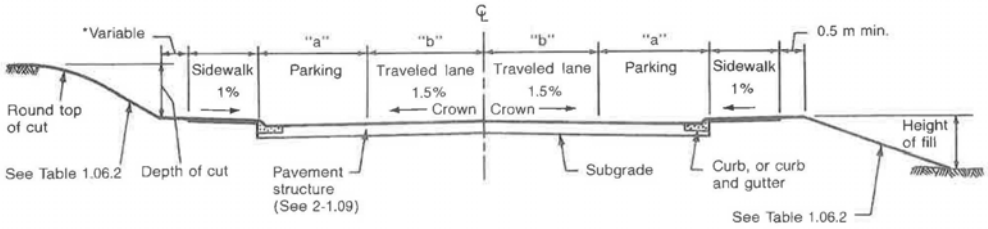
Shoulder and Traveled Lane Widths (m)		
Class	Shoulder "a"	Traveled Lane "b"
Rural Local	2.00	3.25
Rural Collector	2.50	3.65
Rural Arterial	3.00	3.65
Urban Local	2.00	3.25
Urban Collector	2.00	3.65
Urban Arterial	3.00	3.65

See Figure 1.06.5 for Special Treatments

Additional Warrants:

1. For traffic barriers, refer to 2-1.15.
2. For truck climbing lanes, add a width equal to "b".
3. Reduce adjacent "a" to 1.25 m for climbing lane.

Figure 1.06.1
Rural Two-Lane Cross Section
Urban Two-Lane, Noncurbed Cross Section



Parking and Traveled Lane Widths (m)		
Class	Parking "a"	Traveled Lane "b"
Local	2.50	3.25
Collector	2.50	3.65
Arterial	2.00	3.65

*See Figure 1.06.5 for Special Treatments

Figure 1.06.2
Urban Two-Lane Curbed Cross Section

ATTACHMENT 3 VEHICLE DETAILS

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The Assumed Vehicle is 12 Tons Capacity Truck for the FSF Base Camp Road.

Table 1.09.3
Typical*
Truck Class and Weight Distributions

Truck Class	Type SAL-Single Axle TAL-Tandem Axle	Percentage Total Trucks	** Percent of ADT In Design Lane	Gross Vehicle Weight (Ton)	Typical Axle Loads (Tons) EAL Per Axle			EAL Per Vehicle
N.1	Single Unit 2 Axle (2 SAL)	66	14.8	12	2.5		9.5	1.82
					0.01		1.81	
N.2	Single Unit 3 Axle (1 SAL, 1 TAL)	13	2.9	18	3.5		14.5	0.86
					0.03		0.83	
N.3	Multiple Unit 4 Axle (2 SAL, 1 TAL)	11	2.5	21	2.5	8.0	10.5	1.16
					0.01	0.93	0.22	
N.4	Multiple Unit 5 Axle (1 SAL, 2 TAL)	10	2.3	25	3.5	10.0	11.5	0.53
					0.03	0.18	0.32	
Totals/Average		100	22.5					1.48